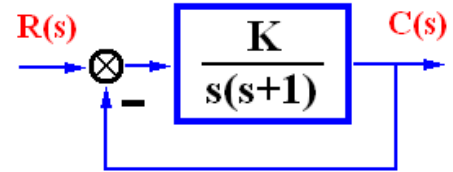


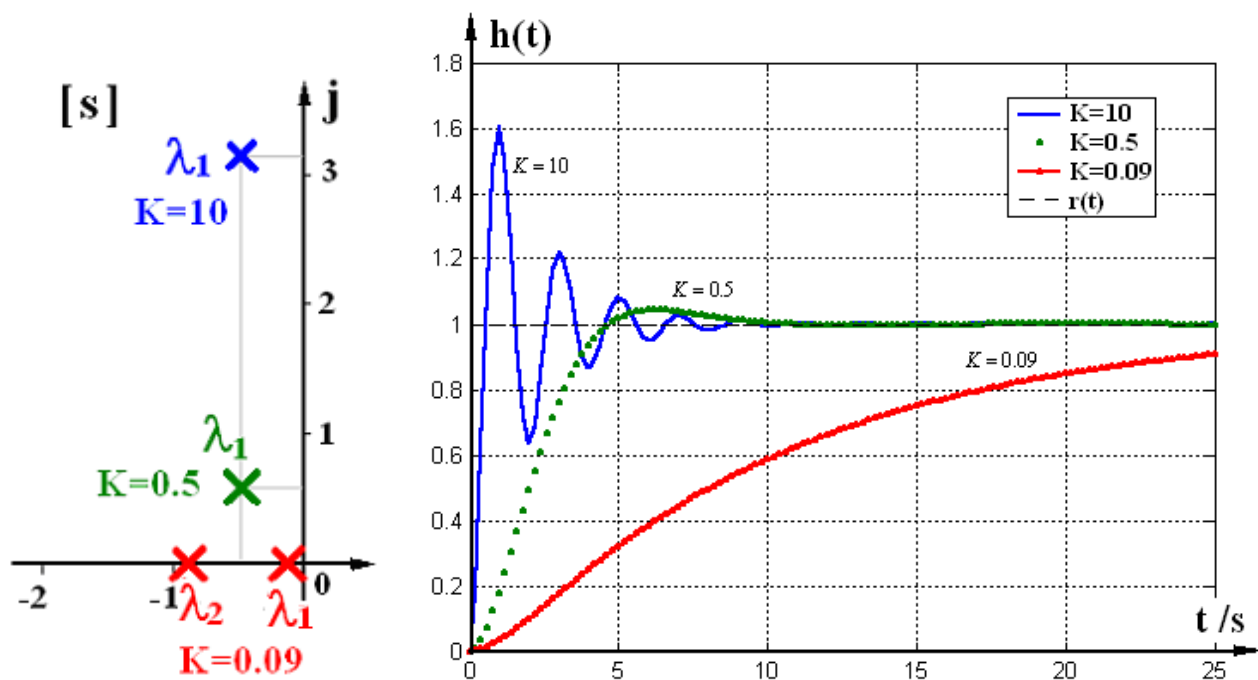
二阶系统性能计算复习

例 系统结构图如图所示。求开环增益 K 分别为 10, 0.5, 0.09 时系统的动态性能指标。



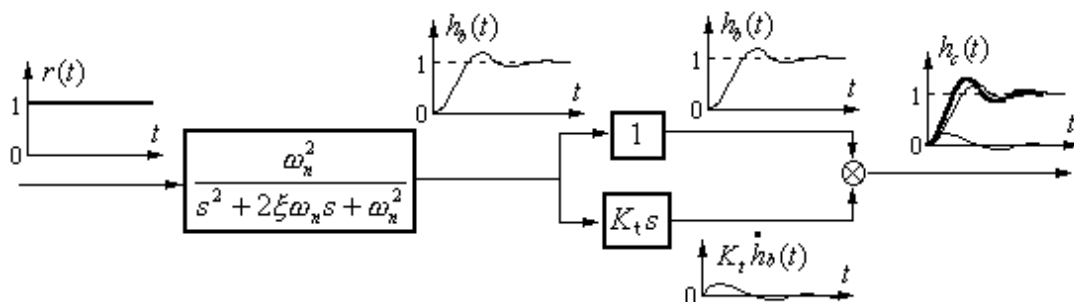
计算过程及结果列表

K 计算	10	0.5	0.09
开环 传递 函数	$G_1(s) = \frac{10}{s(s+1)}$	$G_2(s) = \frac{0.5}{s(s+1)}$	$G_3(s) = \frac{0.09}{s(s+1)}$
闭环 传递 函数	$\Phi_1(s) = \frac{10}{s^2 + s + 10}$	$\Phi_2(s) = \frac{0.5}{s^2 + s + 0.5}$	$\Phi_3(s) = \frac{0.09}{s^2 + s + 0.09}$
特征 参数	$\begin{cases} \omega_n = \sqrt{10} = 3.16 \\ \xi = \frac{1}{2 \times 3.16} = 0.158 \\ \beta = \arccos \xi = 81^\circ \end{cases}$	$\begin{cases} \omega_n = \sqrt{0.5} = 0.707 \\ \xi = \frac{1}{2 \times 0.707} = 0.707 \\ \beta = \arccos \xi = 45^\circ \end{cases}$	$\begin{cases} \omega_n = \sqrt{0.09} = 0.3 \\ \xi = \frac{1}{2 \times 0.3} = 1.67 \end{cases}$
特征 根	$\lambda_{1,2} = -0.5 \pm j3.12$	$\lambda_{1,2} = -0.5 \pm j0.5$	$\begin{cases} \lambda_1 = -0.1 \\ \lambda_2 = -0.9 \end{cases} \begin{cases} T_1 = 10 \\ T_2 = 1.11 \end{cases}$
动态 性能 指标	$\begin{cases} t_p = \frac{\pi}{\sqrt{1-\xi^2} \omega_n} = 1.01 \\ \sigma\% = e^{-\xi\pi/\sqrt{1-\xi^2}} = 60.4\% \\ t_s = \frac{3.5}{\xi\omega_n} = \frac{3.5}{5} = 7 \end{cases}$	$\begin{cases} t_p = \frac{\pi}{\sqrt{1-\xi^2} \omega_n} = 6.238 \\ \sigma\% = e^{-\xi\pi/\sqrt{1-\xi^2}} = 5\% \\ t_s = \frac{3.5}{\xi\omega_n} = 7 \end{cases}$	$\begin{cases} T_1/T_2 = \lambda_2/\lambda_1 = 9 \\ t_s = (t_s/T_1) \cdot T_1 = 31 \\ t_p = \infty, \quad \sigma\% = 0 \end{cases}$

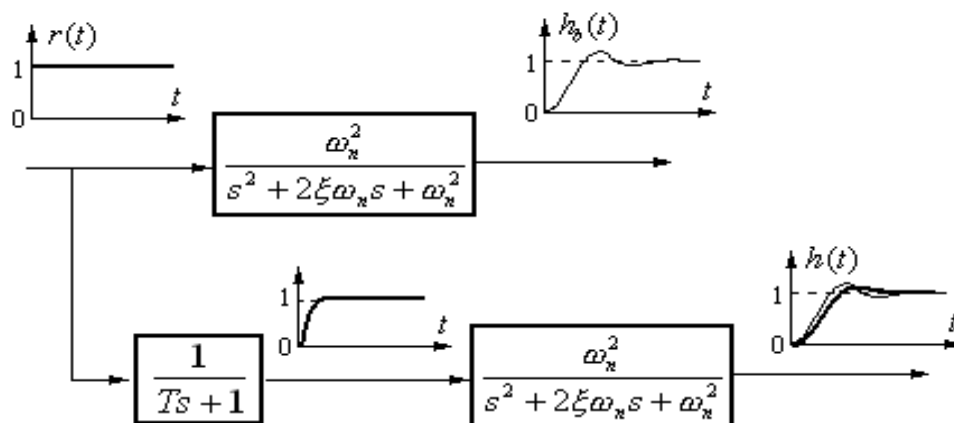


调整参数可以在一定程度上改善系统性能，但改善程度有限

● 附加闭环零/极点对系统性能的影响



附加闭环零点时系统响应合成示意图



附加闭环极点时系统响应合成示意图

§ 3.4 高阶系统的阶跃响应及动态性能

§ 3.4.1 高阶系统单位阶跃响应

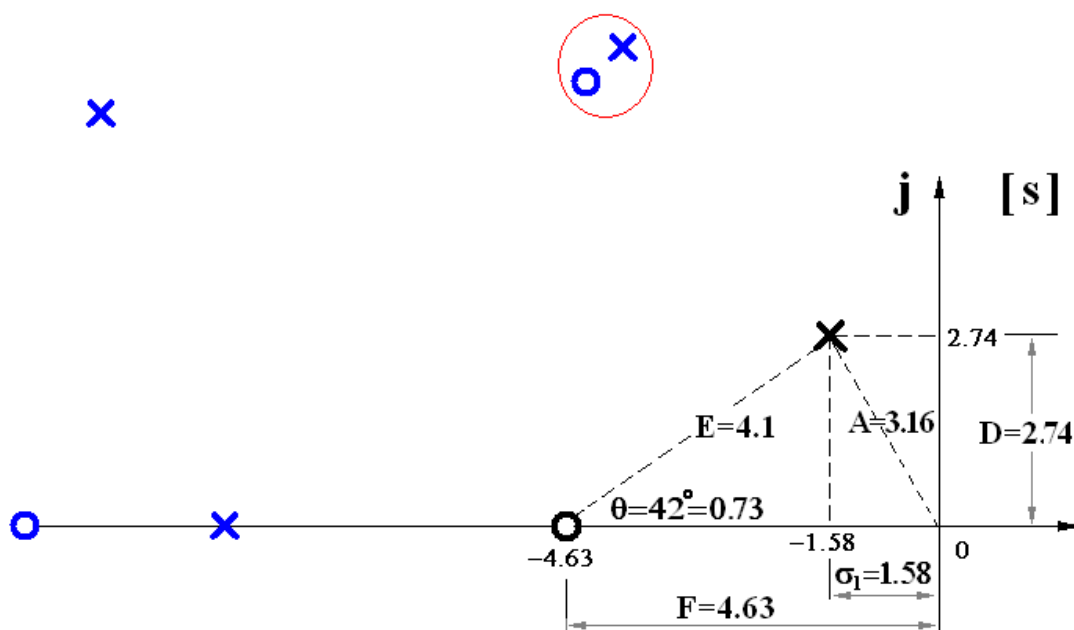
$$\Phi(s) = \frac{M(s)}{D(s)} = \frac{b_m s^m + b_{m-1} s^{m-1} + \cdots + b_1 s + b_0}{a_n s^n + a_{n-1} s^{n-1} + \cdots + a_1 s + a_0}$$

$$= \frac{K \prod_{i=1}^m (s - z_i)}{\prod_{j=1}^n (s - \lambda_j)} \quad n \geq m$$

$$C(s) = \Phi(s) \cdot \frac{1}{s} = \frac{K \prod_{i=1}^m (s - z_i)}{s \prod_{j=1}^n (s - \lambda_j)} = \frac{M(0)}{D(0)} \cdot \frac{1}{s} + \sum_{j=1}^n \frac{M(s)}{s D'(s)} \bigg|_{s=\lambda_j} \frac{1}{s - \lambda_j}$$

$$c(t) = \frac{M(0)}{D(0)} + \sum_{j=1}^n \frac{M(s)}{s D'(s)} \bigg|_{s=\lambda_j} \cdot e^{\lambda_j t}$$

$$= \frac{M(0)}{D(0)} + \sum_{\lambda_i = -\alpha_i} \frac{M(s)}{s D'(s)} \bigg|_{s=\alpha_i} \cdot e^{-\alpha_i t} + \sum_{\lambda_i = -\sigma \pm j\omega_{di}} A_i e^{-\sigma_i t} \sin(\omega_{di} t + \varphi_i)$$



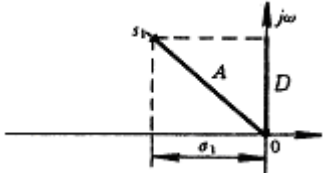
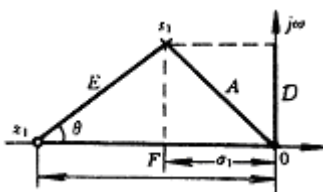
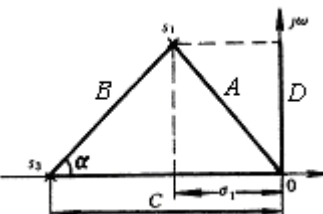
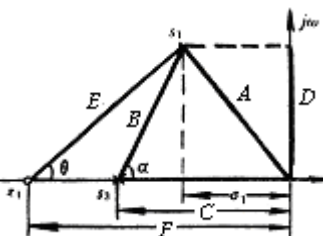
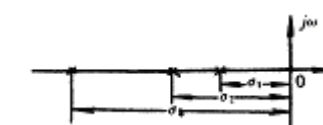
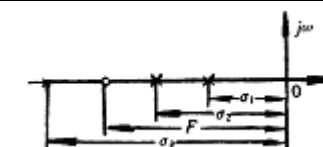
§ 3.4.2 闭环主导极点

主导极点：距离虚轴最近而且附近又没有闭环零点的闭环极点

§ 3.4.3 估算高阶系统动态性能指标的零点极点法

- (1) $\Phi(s) \Rightarrow$ 闭环零极点图；
- (2) 略去非主导零极点和不非常靠近虚轴的“偶极子”，保留主导极点；
- (3) 按 P73 表 3-7 相应公式估算系统动态性能。

表 3-7 动态性能指标估算公式表

系统名称	闭环零、极点分布图	性能指标估算公式
振荡 二阶 系统		$t_p = \frac{\pi}{D}, \quad \sigma\% = 100e^{-\sigma_1 t_p} \%$ $t_s = \frac{3 + \ln\left(\frac{A}{D}\right)}{\sigma_1}$
		$t_p = \frac{\pi - \theta}{D}, \quad \sigma\% = 100 \frac{E}{F} e^{-\sigma_1 t_p} \%$ $t_s = \frac{3 + \ln\left(\frac{A}{D}\right)\left(\frac{E}{F}\right)}{\sigma_1}$
振荡 型三 阶系 统		$t_p = \frac{\alpha}{D}, \quad c_1 = -\left(\frac{A}{B}\right)^2, \quad c_2 = \frac{A}{B} \cdot \frac{C}{D}$ $\sigma\% = 100 \left(\frac{C}{B} e^{-\sigma_1 t_p} + c_1 e^{-c t_p} \right) \%$ $t_s = \frac{3 + \ln c_2}{\sigma_1} \quad (\sigma\% \neq 0 \text{ 时})$ $t_s = \frac{3 + \ln c_1 }{C} \quad (\sigma\% = 0 \text{ 时})$
		$t_p = \frac{\alpha}{D}, \quad c_1 = -\left(\frac{A}{B}\right)^2 \left(1 - \frac{C}{F}\right), \quad c_2 = \frac{A}{B} \cdot \frac{C}{D} \cdot \frac{E}{F}$ $\sigma\% = 100 \left(\frac{C}{B} \cdot \frac{E}{F} e^{-\sigma_1 t_p} + c_1 e^{-c t_p} \right) \%$ $t_s = \frac{3 + \ln c_2}{\sigma_1} \quad (C > \sigma_1, \sigma\% \neq 0 \text{ 时})$ $t_s = \frac{3 + \ln c_1 }{C} \quad (C < \sigma_1, \sigma\% = 0 \text{ 时})$
非振 荡型 三阶 系统		$t_s = \frac{3 - \ln\left(1 - \frac{\sigma_1}{\sigma_2}\right) - \ln\left(1 - \frac{\sigma_1}{\sigma_3}\right)}{\sigma_1} \quad (\sigma_1 \neq \sigma_2 \neq \sigma_3)$
		$t_s = \frac{3 - \ln\left(1 - \frac{\sigma_1}{F}\right) - \ln\left(1 - \frac{\sigma_1}{\sigma_2}\right) - \ln\left(1 - \frac{\sigma_1}{\sigma_3}\right)}{\sigma_1}$ $(\sigma_1 \neq \sigma_2 \neq \sigma_3, F > 1.1\sigma_1 \text{ 时})$

结束

2-15 试绘制图 2-36 所示信号流图对应的系统结构图。

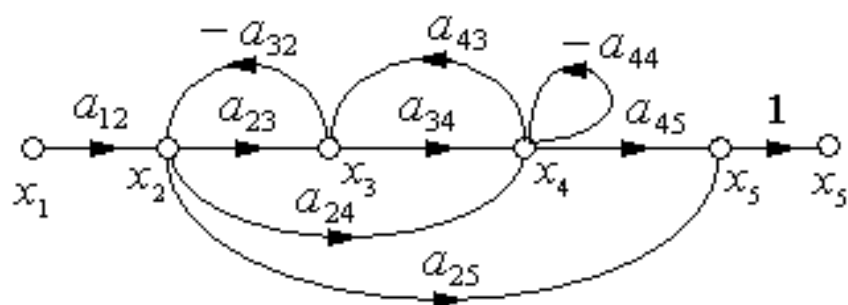
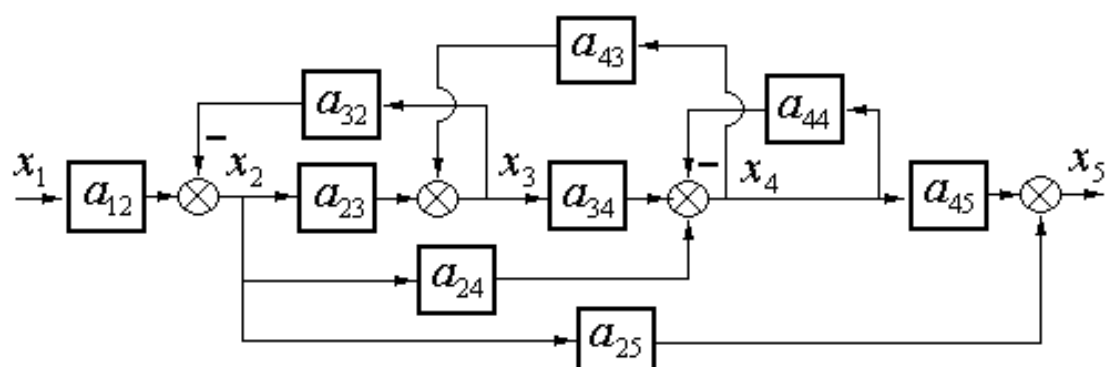
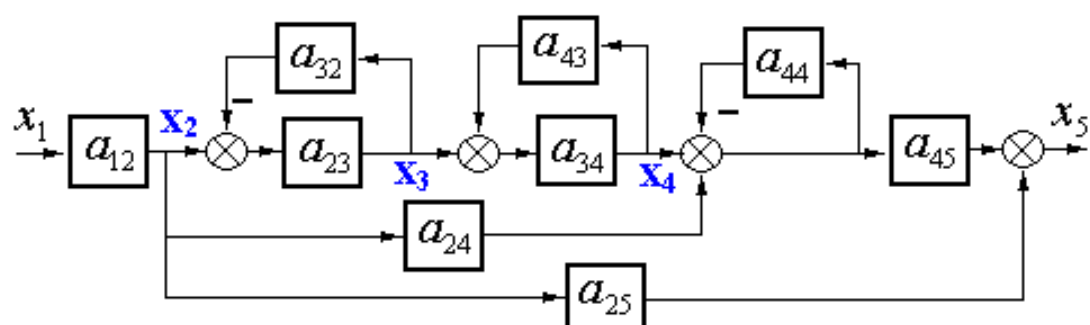


图 2-36

解.



图解2-15

